Portfolio 4 - Heart rate, respiration and interpersonal coordination

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Experimental Methods 3

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Link to code: https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_4_1

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Introduction

(Anders): In this paper an analysis was conducted on the covariance in heart rate and respiration of participants doing a creative LEGO building task. The data was gathered by Fusaroli, R et al. (2016) and consisted of ECG and respiration measurements. Participants were paired and asked to build representations of abstract danish words e.g. *"Justice"*, *"Responsibility"*, *"Knowledge"*.
After preprocessing of the data, the final dataset used in the analysis included data from 4 studies with 6 different conditions and a total of 48 participants paired in 24 unique pairs. The data included 273.350 observations in total.
The data analysis was conducted using R (R Core Team, 2019). The models were built using the packages lme4 (Bates, Maechler, Bolker & Walker, 2015) and lmerTest (Kuznetsova, Brockhoff & Christensen, 2017).

Methods

Data and participants

(Sigrid) Data was collected from undergraduate students of Cognitive Science at Aarhus University, resulting in 97 files in total. Heart rate data (HR) and respiration (resp) data was collected with a frequency of 1000 hertz (measurements / second).
There were a total of 48 unique pairs consisting of 96 participants. Each pair completed 3-5 trials, consisting of 6 conditions: "SelfPaced", "TurnTaking", "Synchronous", "Conversation", "MovementCoop" and "MovementGuided". Conditions were shuffled across trials to keep the order randomized. There were a total of 4 studies.

Preprocessing

(Gustav) The initial data was preprocessed to prepare it for analysis. Before being merged into one data frame each data file was downsampled by taking the mean-value of every 100 rows. Outliers were removed by removing all data points that deviated by more than 2.5 standard deviations from the mean and afterwards scaled and mean-centered. All missing values were removed and the time variable was transformed into sequential data hence we were only interested in the change over time. Figure 1.1 illustrates how different steps of preprocessing changes the data. Note that the downsampled data still looks like the data before it is downsampled.

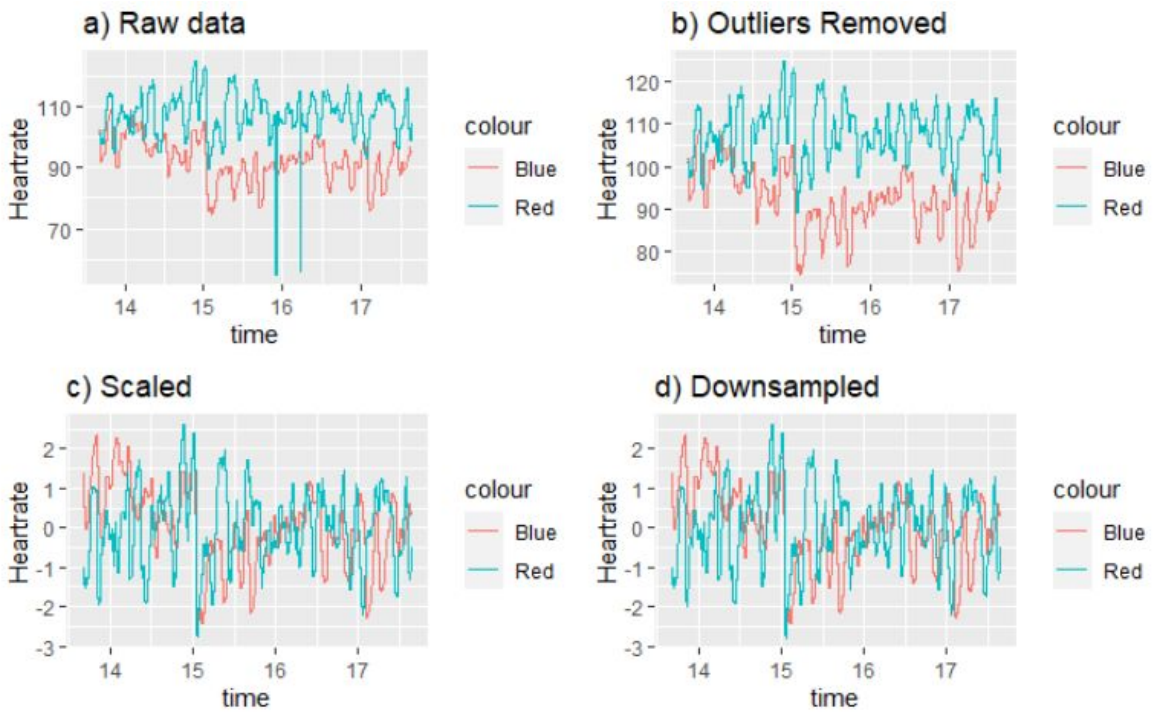

Figure 1.1: Raw data (a), outliers removed (b), scaled data (c) and downsampled data (d). Data is taken from pair1, study 1, trial 1.

Removal of artefacts

(Gustav) Even though outliers above 2.5 or below -2.5 standard deviations were removed, artifacts were still present in the data. We searched for artefacts by plotting the heart rate and the respiration data for each participant, respectively. We excluded files by eyeballing the data, trying to detect noisy data. 18 files were excluded, and the names of these can be found in table 1 in appendix A. An example of both included and excluded files for both heart rate and respiration data can be found in figure 1.2.

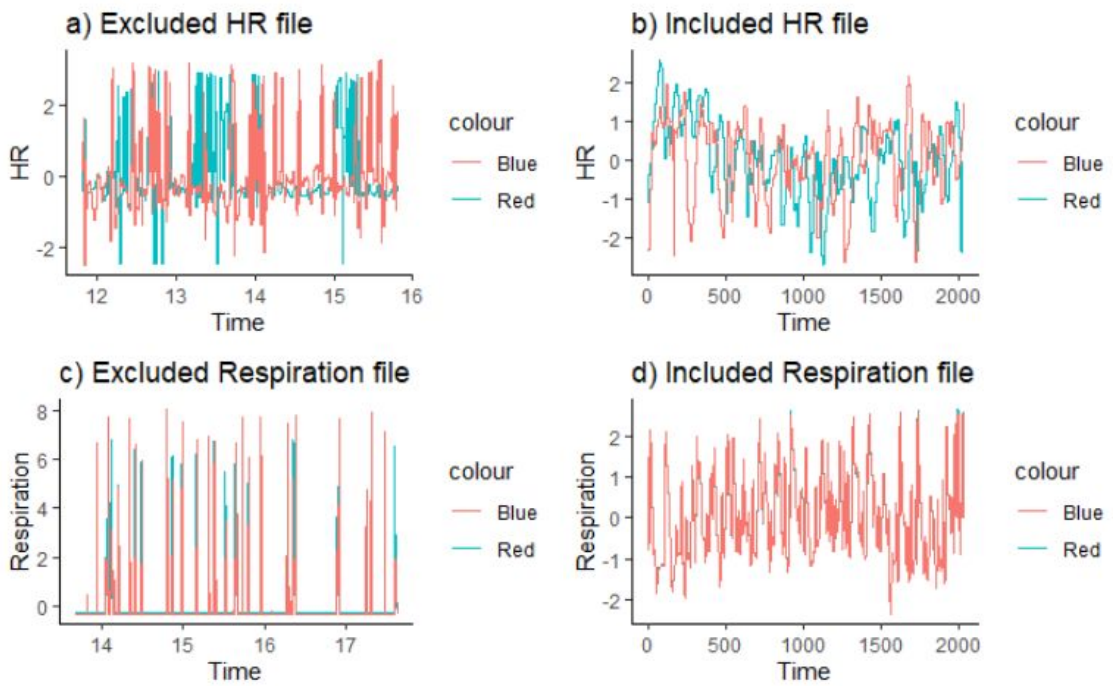

Figure 1.2: Excluded HR file (a), included HR file (b), excluded respiration file (c) and included respiration file (d).

Variables and models

(Anders) The models used in the analysis were designed to predict the change of either the respiration or HR data as a function of the first participant's data and the difference between the first participant's data and the data of the second participant. The models approximated the differential equation models proposed in Ferrer & Helm (2013). By assuming that the current change rate between each data point is equal to a differential term ($\frac{dx}{dt}$) we can use a differential equation as a model. The frequency of our measures is 10 Hz, though not instantaneous, it is a justifiable representation of the original differential term. For a more thorough review of the model design see Ferrer & Helm (2013).

The general differential equation depicted in Ferrer & Helm (2013) is written as follows:

$$\frac{dx}{dt} = a_1 * x_t + a_2(y_t - x_t) \qquad (1)$$

(Gustav) We were interested in modelling change in heart rate and respiration. We defined change as the difference between one's future heart rate (i.e. at the next time-point) and one's current heart rate. Two variables were used to predict heart rate: stability and coupling. Stability is a measure of how much the participant's heart rate is affected by her own heart rate. Coupling is a measure of how much the participant's heart rate is affected by the heart rate of the partner. These correspond to x_t and $(y_t - x_t)$ as seen in (1), respectively. After running the model we get two alpha-estimates for each condition, a_1 and a_2. Table 1.1 shows the interpretation of a_1 and a_2 given that they are significant.

	a_1	a_2
Positive	Self-instability away from homeostasis	Signal attracted to partner
Negative	Self-regulation towards homeostasis	Signal repelled from partner

Table 1.1: Interpretation of a_1 and a_2 of the model.

(Sigrid) We are interested in modelling the interactions between conditions and thereby investigate whether the different conditions elicited different tendencies of covariation. We thereby only included interaction effects in the model and left out fixed effects (thus typing ‘ : ’ rather than ‘ * ‘ in the model). None of the models included intercepts hence we were only interested in modelling the change which made the intercept redundant (the ‘0’ in the model). To meet the assumption of independence, random intercepts were added for participant, pair and study and for each of these random intercepts, a random slope for condition was added. “HR_self” was entered as a fixed effect to model stability within a participant, which corresponds to a_1. “HR_other - HR_self” was entered as a fixed effect to model coupling within the pair, which corresponds to a_2. The ‘Type’ interaction is a control condition which will be further explained in the “Control Conditions” section.

The models used in the analysis were written using the following syntax:

Heart rate model:

lmer(HR_change ~ 0 + (HR_self + (HR_other - HR_self) : Condition : Type + (0 + Condition|Participant) + (0 + Condition|Pair) + (0 + Condition|Study))

Respiration rate model:

lmer(Resp_change ~ 0 + (Resp_self + (Resp_other - Resp_self) : Condition : Type + (0 + Condition|Participant) + (0 + Condition|Pair) + (0 + Condition|Study))

The models will elicit four different outputs due to ‘Type’ being two different control conditions. This will be elaborated in the following sections.

Control conditions

(Sigrid) In order to investigate whether it is the two participants conducting the experiment who affect each other or if it is merely the condition itself, the model was run on a data set consisting of both real data and a control condition (see the 'type' predictor in the model). Thus, if the hypotheses of this experiment were to not be rejected, only the interaction effects with real data should be found to significantly predict change in heart rate. If significant results are found within both the real data and the control condition, it places mistrust in the conclusions of the significant real data. That is, the chosen hypotheses can only be supported by significant results of the real data, if the control data simultaneously produced nonsignificant results.
We created two control conditions; one with shuffled controls and one with surrogate pairs. This will result in four models total, two for each type of control condition.

Shuffled controls

(Sigrid) A new dataset was created by sampling all of the predictor variables using the sample() function in R (R Core Team, 2019). The function randomizes the values within each of the columns. That is, it does not create new values, it merely swaps the placement of the values within each column. That way a randomized dataset was created and used to predict the change in heart rate alongside the real data.

Surrogate pairs

(Gustav) The surrogate pairs were made by replacing the second participant of each pair with another second participant from another pair. Participants were only shuffled within study and condition. This made sure that the new pairs would always consist of data from the same study and condition hereby making sure that too much unaccounted variance were not added to the pairs. The new surrogate pair labels were then added to the data frame and the "HR_other" and "Resp_other" variable was updated to fit the data of the other participant in the surrogate pair. This dataset was then combined with the original data and the "Type" column was added in order to differentiate between real pairs and surrogate pairs.

Results

Plots according to condition

(Gustav) A plot depicting the heart rate data for one pair in each condition can be found in figure 2.1

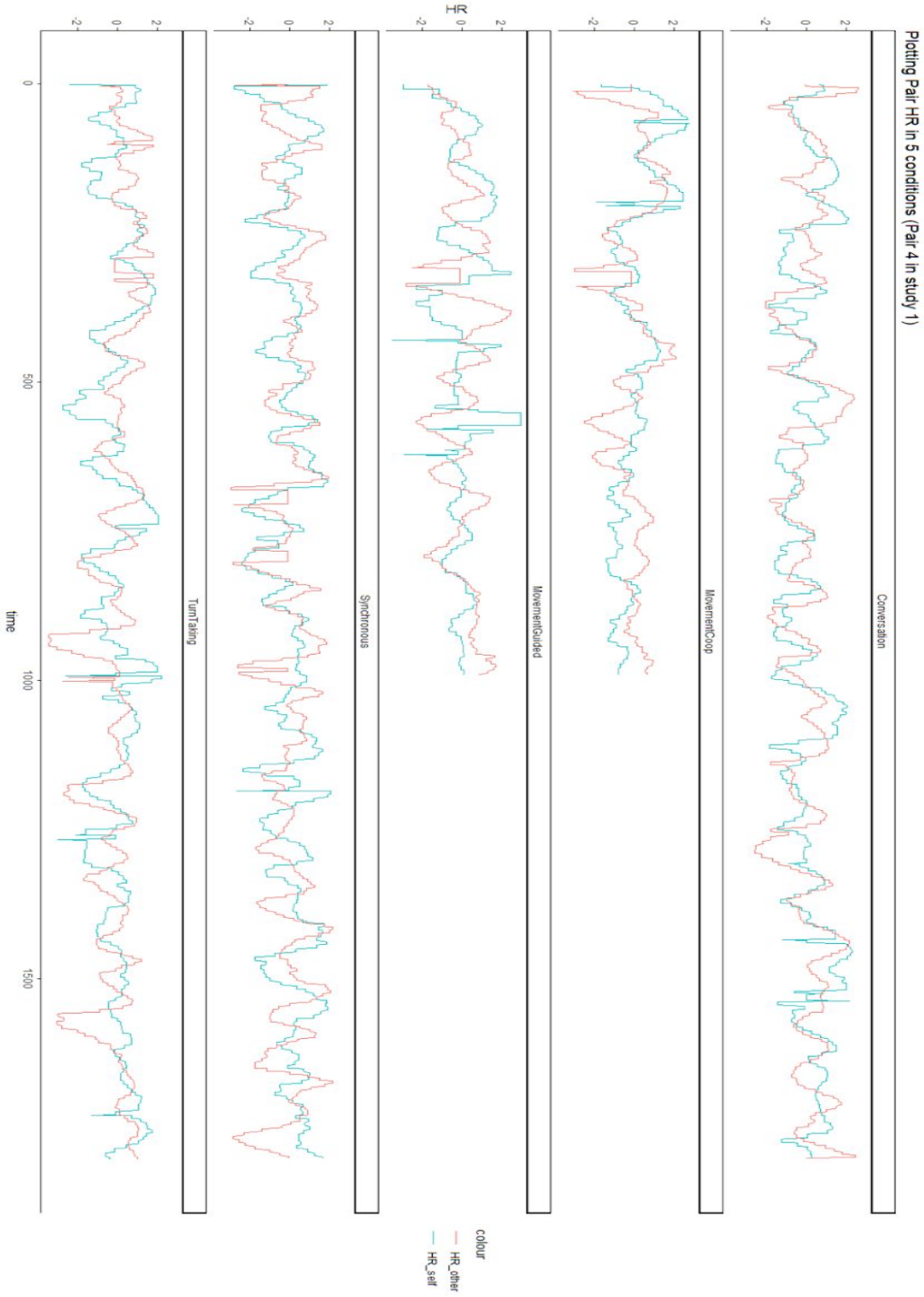

Figure 2.1: Plot depicting heart rate in the 6 different conditions for one pair

A plot depicting the respiration data for one pair in each condition can be found in figure 2.2.

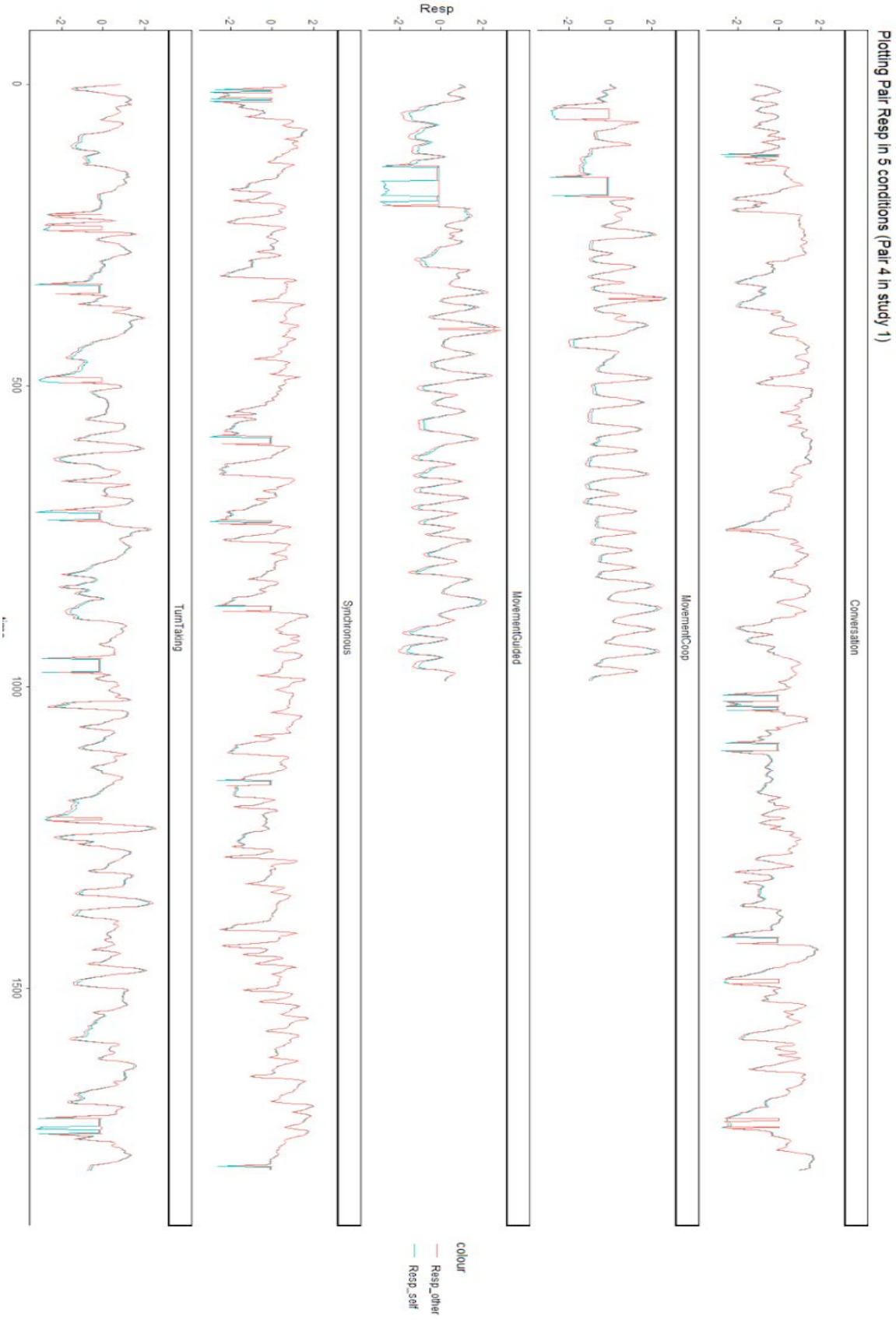

Figure 2.2: Plot depicting respiration in the 6 different conditions for one pair.

Real data and control conditions

(Anders) The tables found below display the fixed effects outputs for the four models depicted in the "Methods" section.

The marking of the p-values is as following: . = p < .1, * = p < .05, ** = p < .01, *** = p < .001

Shuffled controls for heart rate

The results of the model containing the shuffled controls for heart rate can be found in the table 2.1 below. All the below mentioned effects have 5.467e+05 degrees of freedom.

Fixed effect (interaction)	**β-estimate**	**SE**	**t-value**	**p-value**
HR_SELF real data (alpha 1)				
HR_self:Conversation:Real	-3.25e-02	9.84e-04	-33.06	<.0001 ***
HR_self:MovementCoop:Real	-3.67e-02	2.02e-03	-18.18	<.0001 ***
HR_self:MovementGuided:Real	-5.04e-02	2.00e-02	-25.15	<.0001 ***
HR_self:SelfPaced:Real	-3.47e-02	2.19e-03	15.88	<.0001 ***
HR_self:Synchronous:Real	-3.77e-02	9.28e-04	-40.64	<.0001 ***
HR_self_TurnTaking:Real	-2.85e-02	9.49-04	-30.08	<.0001 ***
HR_SELF shuffled data (alpha 1)				
HR_self:Conversation:Shuffled	-6.18e-04	9.83e-04	-0.63	0.53
HR_self:MovementCoop:Shuffled	-5.50e-03	1.97e-03	-2.791	0.0053 **
HR_self:MovementGuided:Shuffled	1.62e-03	2.00e-03	0.81	0.42
HR_self:SelfPaced:Shuffled	-4.14e-04	2.14e-03	-0.194	0.85
HR_self:Synchronous:Shuffled	-6.32e-04	9.21e-04	-0.686	0.49
HR_self:TurnTaking:Shuffled	-2.36e-04	9.44e-04	-0.250	0.80
HR_OTHER real data (alpha 2)				
HR_other:Conversation:Real	1.42e-03	9.84e-04	1.439	0.15

HR_other:MovementCoop:Real	7.27e-03	2.02e-03	3.61	0.00031***
HR_other:MovementGuided:Real	-2.959e-03	2.00e-03	-1.48	0.14
HR_other:SelfPaced:Real	7.82e-03	2.19e-03	3.58	0.00035*
HR_other:Synchronous:Real	4.78e-03	9.28e-04	5.15	<.0001***
HR_other:TurnTaking:Real	-2.87e-03	9.49e-04	-3.026	0.0025**
HR_OTHER shuffled data (alpha 2)				
HR_other:Conversation:Shuffled	1.17e-03	9.83e-04	1.190	0.23
HR_other:MovementCoop:Shuffled	-3.88e-03	1.97e-03	-1.967	0.049*
HR_other:MovementGuided:Shuffled	1.83e-03	2.00e-03	0.915	0.36
HR_other:SelfPaced:Shuffled	-7.46e-03	2.14e-03	-0.349	0.73
HR_other:Synchronous:Shuffled	2.63e-04	9.21e-04	0.286	0.77
HR_other:TurnTaking:Shuffled	2.99e-04	9.44e-04	0.316	0.75

Table 2.1: Results of both real and shuffled heart rate data. .

Shuffled controls for respiration

The results of the model containing the shuffled controls for respiration can be found in the table 2.2 below. All the below mentioned effects have 5.467e+05 degrees of freedom.

Fixed effect (interaction)	**β-estimate**	**SE**	**t-value**	**p-value**
Resp_SELF real data (alpha 1)				
Resp_self:Conversation:Real	-2.94e-01	3.71e-03	-79.46	<.0001***
Resp_self:Movement:Real	-1.95e-01	6.77e-03	-28.712	<.0001***
Resp_self:MovementGuided:Real	-2.07e-01	7.54e-03	-27.392	<.0001***
Resp_self:SelfPaced:Real	-1.67e-01	5.29e-03	-31.58	<.0001***
Resp_self:Synchronous:Real	-1.99e-01	2.83-03	-70.348	<.0001***

Resp_self:TurnTaking:Real	-2.58e-01	3.98e-03	-64.88	<.0001***
Resp_SELF shuffled data (alpha 1)				
Resp_self:Conversation:Shuffled	2.69e-04	1.03e-03	0.26	0.79
Resp_self:Movement:Shuffled	4.76e-03	2.07e-03	2.30	0.0213*
Resp_self:MovementGuided:Shuffled	-5.13e-04	2.10e-03	-0.25	0.81
Resp_self:SelfPaced:Shuffled	-5.59e-03	2.24e-03	-2.50	0.0126*
Resp_self:Synchronous:Shuffled	-9.53e-04	0.65e-04	-0.99	0.32
Resp_self:TurnTaking:Shuffled	1.86-04	9.89e-04	0.19	0.85
Resp_OTHER real data (alpha 2)				
Resp_other:Conversation:Real	2.65e-01	3.71-03	71.62	<.0001***
Resp_other:MovementCoop:Real	1.63e-01	6.77e-03	23.99	<.0001***
Resp_other:MovementGuided:Real	1.75e-01	7.54e-03	23.21	<.0001***
Resp_other:SelfPaced:Real	1.36e-01	5.29e-03	25.64	<.0001***
Resp_other:Synchronous:Real	1.69e-01	2.83e-03	59.84	<.0001***
Resp_other:TurnTaking:Real	2.31e-01	3.98e-03	58.20	<.0001***
Resp_OTHER shuffled data (alpha 2)				
Resp_other:Conversation:Shuffled	6.33e-04	1.03e-03	0.62	0.54
Resp_other:MovementCoop:Shuffled	1.36e-03	2.07e-03	0.66	0.51
Resp_other:MovementGuided:Shuffled	-3.44e-03	2.10e-03	-1.64	0.10
Resp_other:SelfPaced:Shuffled	2.11e-03	2.24e-03	0.94	0.35
Resp_other:Synchronous:Shuffled	-7.14e-04	9.65e-04	-0.74	0.46
Resp_other:TurnTaking:Shuffled	-1.79e-04	9.89e-04	-0.18	0.86

Table 2.2: Results of both real and shuffled respiration data.

Surrogate pair controls for heart rate

The results of the model containing the surrogate pair controls for heart rate can be found in the table 2.3 below. All the below mentioned effects have 4.9e+05 degrees of freedom.

Fixed effect (interaction)	**β-estimate**	**SE**	**t-value**	**p-value**
HR_SELF real data (alpha 1)				
HR_self:Conversation:Real	-3.52e-02	9.90e-04	-32.82	<.0001 ***
HR_self:MovementCoop:Real	-3.67e-02	2.03e-03	-18.07	<.0001 ***
HR_self:MovementGuided:Real	-5.04e-02	2.02e-03	-24.99	<.0001 ***
HR_self:SelfPaced:Real	-3.47e-02	2.20e-03	-15.78	<.0001 ***
HR_self:Synchronous:Real	-3.77e-02	9.34e-04	-40.39	<.0001 ***
HR_self_TurnTaking:Real	-2.85e-02	9.54e-04	-29.90	<.0001 ***
HR_SELF surrogate pairs (alpha 1)				
HR_self:Conversation:Surrogate	-3.41e-02	1.06e-03	-32.11	<.0001 ***
HR_self:MovementCoop:Surrogate	-3.47e-0.2	2.13e-03	-16.31	<.0001 ***
HR_self:MovementGuided:Surrogate	-4.77e-0.2	2.13e-03	-22.41	<.0001 ***
HR_self:SelfPaced:Surrogate	-3.44e-02	2.24e-03	-15.33	<.0001 ***
HR_self:Synchronous:Surrogate	-3.94e-02	1.02e-03	-38.54	<.0001 ***
HR_self:TurnTaking:Surrogate	-3.04e-02	1.14e-03	-26.64	<.0001 ***
HR_OTHER real data (alpha 2)				
HR_other:Conversation:Real	1.42e-03	9.90e-04	1.43	0.15
HR_other:MovementCoop:Real	7.27e-03	2.03e-03	3.58	0.003 ***
HR_other:MovementGuided:Real	-2.96e-03	2.02e-03	-1.47	0.14
HR_other:SelfPaced:Real	7.82e-03	2.20e-03	3.55	0.0003 ***
HR_other:Synchronous:Real	4.78e-03	9.34e-04	5.12	<.0001 ***

HR_other:TurnTaking:Real	-2.87e-03	9.54e-04	-3.01	0.0026 **
HR_OTHER surrogate pair (alpha 2)				
HR_other:Conversation:Surrogate	1.22e-03	1.06e-03	1.15	0.25
HR_other:MovementCoop:Surrogate	2.82e-03	2.13e-03	1.33	0.18
HR_other:MovementGuided:Surrogate	9.02e-04	2.13e-03	0.42	0.67
HR_other:SelfPaced:Surrogate	1.15e-03	2.24e-03	0.51	0.61
HR_other:Synchronous:Surrogate	1.52e-03	1.02e-03	1.49	0.14
HR_other:TurnTaking:Surrogate	-5.31e-04	1.14e-03	-0.47	0.64

Table 2.3: Results of both real and surrogate pairs heart rate data.

Surrogate pair controls for respiration

The results of the model containing the surrogate pair controls for respiration can be found in the table 2.4 below. All the below mentioned effects have 4.948e+05 degrees of freedom.

Fixed effect (interaction)	**β-estimate**	**SE**	**t-value**	**p-value**
Resp_SELF real data (alpha 1)				
Resp_self:Conversation:Real	-2.94e-01	3.65e-03	-80.60	<.0001***
Resp_self:MovementCoop:Real	-1.95e-01	6.68e-03	-29.12	<.0001***
Resp_self:MovementGuided:Real	-2.06e-01	7.43e-03	-27.78	<.0001***
Resp_self:SelfPaced:Real	-1.67e-01	5.22e-03	-32.02	<.0001***
Resp_self:Synchronous:Real	-1.99e-01	2.79e-03	-71.35	<.0001***
Resp_self_TurnTaking:Real	-2.58e-01	3.92e-03	-65.80	<.0001***
Resp_SELF surrogate data (alpha 1)				
Resp_self:Conversation:Surrogate	-3.71e-02	1-08e-03	-34.12	<.0001***

Resp_self:MovementCoop:Surrogate	-3.36e-02	2.17e-03	-15.50	<.0001***
Resp_self:MovementGuided:Surrogate	-3.76e-02	2.22e-03	-16.94	<.0001***
Resp_self:SelfPaced:Surrogate	-4-57e-02	2.37e-03	-19.26	<.0001***
Resp_self:Synchronous:Surrogate	-3.83e-02	1.05e-03	-36.38	<.0001***
Resp_self:TurnTaking:Surrogate	-3.48e-02	1.17e-03	-29.7	<.0001***
Resp_OTHER real data (alpha 2)				
Resp_other:Conversation:Real	2.65e-01	3.65e-03	72.64	<.0001***
Resp_other:MovementCoop:Real	1.63e-01	6.67e-03	24.33	<.0001***
Resp_other:MovementGuided:Real	1.75e-01	7.43e-03	23.53	<.0001***
Resp_other:SelfPaced:Real	1.40e-01	5.21e-03	26.00	<.0001***
Resp_other:Synchronous:Real	1.69e-01	2.78e-03	60.69	<.0001***
Resp_other:TurnTaking:Real	2.314e-01	3.92e-03	59.03	<.0001***
Resp_OTHER surrogate data (alpha 2)				
Resp_other:Conversation:Surrogate	6.16e-04	1.09e-03	0.57	0.57
Resp_other:MovementCoop:Surrogate	-7.36e-04	2.17e-03	-0.34	0.73
Resp_other:MovementGuided:Surrogate	-3.95e-04	2.21e-03	-1.78	0.075 .
Resp_other:SelfPaced:Surrogate	1.11e-03	2.37e-03	0.47	0.63
Resp_other:Synchronous:Surrogate	2.68e-03	1.05e-03	2.54	<0.5*
Resp_other:TurnTaking:Surrogate	5.36e-04	1-17e-03	0.46	0.647

Table 2.4: Results of both real and surrogate pairs for the respiration data.

Discussion

Two different control conditions serve two different purposes

(Gustav) Two control conditions were used in the analysis: shuffled controls and surrogate pairs.
In the data where shuffled controls were used, all the data for HR_self, HR_other, Resp_self and Resp_other have been randomly shuffled. In the following discussion, the shuffled control condition is used to validate or invalidate the significant results of only HR_self and Resp_Self (a_1).
The other control condition, surrogate pairs, assigns a new pair to each participant and thus changes the values HR_other and Resp_other while keeping the values of HR_self and Resp_self constant. Thus, the surrogate pair control condition is used to validate or invalidate the significant results for only HR_other and Resp_other (a_2).
We could also have chosen to use shuffled controls as an extra control condition for HR_other and Resp_other. However, we think that the surrogate pairs condition is more controlled (conforms to study and condition) and thereby better suited as a control condition for HR_other and Resp_other.

Heart rate

Shuffled Controls

(Anders) HR_self was found to be significant across all conditions in the real data (table 2.1). This indicates a stability within each individual's heart rate. However, HR_self was also significant in the shuffled data for the condition “MovementCoop”. This thereby invalidates the significant result for HR_self in the same condition. In the remaining 5 conditions, all alpha-estimates are negative, meaning that participants are moving toward homeostasis. Thus, using shuffled controls as a control condition, we find a significant effect of stability in 5 out of 6 of the conditions (excluding “MovementCoop”).

Surrogate Pairs

(Anders) All the results for the real data for HR_self are the same as mentioned above in the “Shuffled controls”, in terms of significance and alpha estimates. In the surrogate pair data we also find significant results across all conditions for HR_self (table 2.3). This does not invalidate the significant results for HR_self in the real data, because the surrogate data only changes the values of HR_other, not HR_self.

We can use surrogate pairs to validate or invalidate HR_other.
HR_other was found to be significant in 4 out of 6 conditions in the real data. The significant conditions were: “MovementCoop”, “SelfPaced”, “Synchronous” and “Turntaking”.

HR_other was not found to be significant in any of the conditions in the surrogate pair data. Thus, using surrogate pairs as a control condition for HR_other, we find that it is significant across all 4 conditions. In the 4 conditions the alpha values were positive for “MovementCoop”, “SelfPaced” and “Synchronous” while being negative for “Turntaking”. Thus, we find a significant effect of coupling *towards* the partner in the conditions “MovementCoop”, “Selfpaced” and “Synchronous”. In contrast, there is a significant effect of coupling *away* from the partner in “TurnTaking”.

Respiration

Shuffled Controls

(Sigrid) The real data for Resp_self were in general found to be significant, while the shuffled data were not, apart from two instances of Resp_self being significant even though the data were shuffled (table 2.2). These two instances were, however, significant at a higher threshold (p = <. 05) than the real data (p = <.0001). Though these two instances could be due to chance, we cannot say anything with certainty. This discredits the significant results of Resp_self in the two conditions in question, which are ‘MovementCooperated’ and ‘SelfPaced’.
If we set aside the two two incidents of significant results of the shuffled data, the results in general indicate that the respiration of the participant in question seems to remain stable, that is, it seems to go towards homeostasis because of the negative alpha values.

Surrogate Pairs

(Sigrid) As mentioned, it is only the effect of Resp_other (a_2) that is relevant in the surrogate pair control condition. All but one of the effects of the surrogate pair data were found to be non-significant, while all of the real pairs were found to be significant (table 2.4). The only significant surrogate pair effect was found in the ‘Synchronous’ condition (p = <. 05). Although the effects of the real data were significant at a lower level (p = <.0001), the significance of the surrogate pair effect still discredits the significant results of Resp_other in the mentioned condition in the real data.
The effects of Resp_other in the remaining conditions for the real data all indicate that the respiration of the participant in question aligns with the respiration of the other participant.

Summary of interpreted results

The following table provides a summary for all the significant results (3.1).

		Significant conditions	Alpha direction	Effect sizes
Heart Rate	HR_self stability (a_1)	Conversation	stability	-3.25e-02
		Movement Guided	stability	-5.04e-02
		Self Paced	stability	-3.47e-02
		Synchronous	stability	-3.77e-02
		Turn Taking	stability	-2.85e-02
	HR_other coupling (a_2)	Movement Coop	coupling	7.27e-03
		Self Paced	coupling	7.82e-03
		Synchronous	coupling	4.78e-03
		Turn Taking	**decoupling**	-2.87e-03
Respiration	Resp_self stability (a_1)	Conversation	stability	-2.94e-01
		Movement Guided	stability	-2.07e-01
		Synchronous	stability	-1.99e-01
		Turn Taking	stability	-2.58e-01
	Resp_other coupling (a_2)	Conversation	coupling	2.65e-01
		Movement Coop	coupling	1.63e-01
		Movement Guided	coupling	1.75e-01
		Self Paced	coupling	1.36e-01

		Turn Taking	coupling	2.31e-01

Table 3.1: Summary of the significant scaled effects size for HR and respiration data across all significant conditions.

(Anders) We found an effect of stability in both heart rate and respiration in the following 4 conditions: Conversation, Movement Guided, Synchronous and Turn Taking.
We found an effect of coupling within pairs in both heart rate and respiration in two conditions: Movement Coop and Self paced. The condition "Turn Taking" produced significant results for both heart rate and respiration. However, in the respiration data it was found to produce coupling, whereas in the heart rate data it was found to produce a decoupling effect. This is a contradictory result.

In general, it seems that the scaled effect sizes are larger for the respiration data compared to the heart rate data. This will be elaborated in the last section of this assignment.

(Sigrid) The largest scaled effects of the respiration data are observed in the 'Conversation' and the 'TurnTaking' condition for both Resp_self and Resp_other. In Resp_self the difference between the largest and the smallest scaled effect size is 0.01, while in Resp_other the difference between the largest and the smallest effect size is 0.13. Thus, the difference between the scaled effect sizes of Resp_self is fairly small. That is, the four significant conditions do have an effect on the stability of the participant's respiration, however, the effect does not vary a lot within the four conditions.
The variation of the effect sizes for Resp_other does, however, seem to vary a bit more when compared to Resp_self. A variation of 0.13 between effect sizes could be concluded to be a relatively big effect, since the effect sizes in general are quite small.

The variance of the scaled effect sizes for stability and coupling in heart rate is relatively small. The conditions yield effect sizes that are quite similar to each other. Thus, there is a low variability of effect sizes within conditions.

Is respiration coordination a likely driver of heart coordination?

(Anders) The analysis showed a significant covariation effect for both respiration and heart rate between participants within pairs. This initially leads to the question of whether respiration and heart rate internally are correlated and if so, whether participants in reality only align to each other on one of the two parameters. In that case, the other parameter shows a significant covariation effect as a consequence to its correlation to the aligned parameter. The next sections will be discussing the causation of the significant effects found in our analysis.

(Sigrid) Our results showed that the covariation effect of respiration between participants was stronger than the covariation effect of heart rate (see figure 1.2b and 1.2d). If heart rate is dependent on respiration, it could possibly be explained by the fact that participants only aligned their respiration to the respiration of their partner and not their heart rate. However, since heart rate is correlated with respiration, the heart rate of the two participants, respectively, becomes synchronous as a consequence of this. This could be tested by introducing two new hypotheses; H1) Heart rate is predicted by respiration, and H2) Participants align their respiration to the respiration of their partner, but not their heart rate.

(Gustav) To test H1, one could simply run a model that tested whether heart rate was predicted by respiration. If respiration significantly predicts heart rate, it could be argued that since the covariation effect is much more apparent in the respiration data than in the heart rate data, the covariation effect found in heart rate data is a secondary effect of respiration rather than an alignment to the partner's heart rate. A possible explanation to this could be that respiration is a process that is easy to detect externally both visually and auditorily in bodily motions and sounds associated with respiration. Conversely, it is much more difficult to detect heart rate only by visual and auditory cues. However, Li, Chen, Zhao & Pietikäinen (2014) have been able to successfully detect heart rate only by visual cues from video data of faces. If a machine is able to detect heart rate only by visual cues why should humans not be able to do it as well? In order to determine the two parameters' true covariation effect a new study design is needed that can make an accurate dissociation between respiration and heart rate covariation. The question is if this is possible since it is hard to imagine two tests that separate respiration and heart rate since they will always be interconnected in the human system.

(Gustav) The results of our study show that a covariation effect is present. Participants did align both their respiration and heart rate to that of their partner. However, we are not able to determine the internal relationship between heart rate and respiration. We are also not able to determine whether the covariation effect found for heart rate and respiration are a true effect or a secondary effect of the alignment to the other parameter i.e. we cannot determine the dependency of the two parameters.

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Appendix A

The following 18 files were removed due to artefacts in either the heart rate and/or the respiration data. Each row corresponds to one file.

Study	Pair	Trial
1	1	1
1	1	2
1	3	2
2	5	1
2	5	2
2	5	3
2	6	1
2	6	2
2	6	3
2	8	1
2	8	2
2	8	3
3	1	3
3	9	2
4	1	4
4	3	3
4	3	5
4	6	1

Table 1: The 18 files which were excluded due to artefacts. Each row corresponds to one file.